



Chapter 2

Installing and Managing Windows Server 2022

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August, 2024

Content

- Requirements for installation
- Installing Windows Server 2022
- Installing roles and features
- Centralized management and monitoring
- Windows Admin Center (WAC)
- Enabling quick server rollouts with Sysprep
- In-place upgrading to Windows Server 2022



Technical requirements

- Many of the decisions that you need to make are licensing-type decisions
- What roles do you intend to install on this server?
- Do they require a special edition?
- Can the more common Server 2022 Standard edition handle it, or do we need the Datacenter edition for our purposes?
- Is Server Core going to be beneficial from a security perspective, or do we need the full Desktop Experience?



Technical requirements

Minimum system requirements

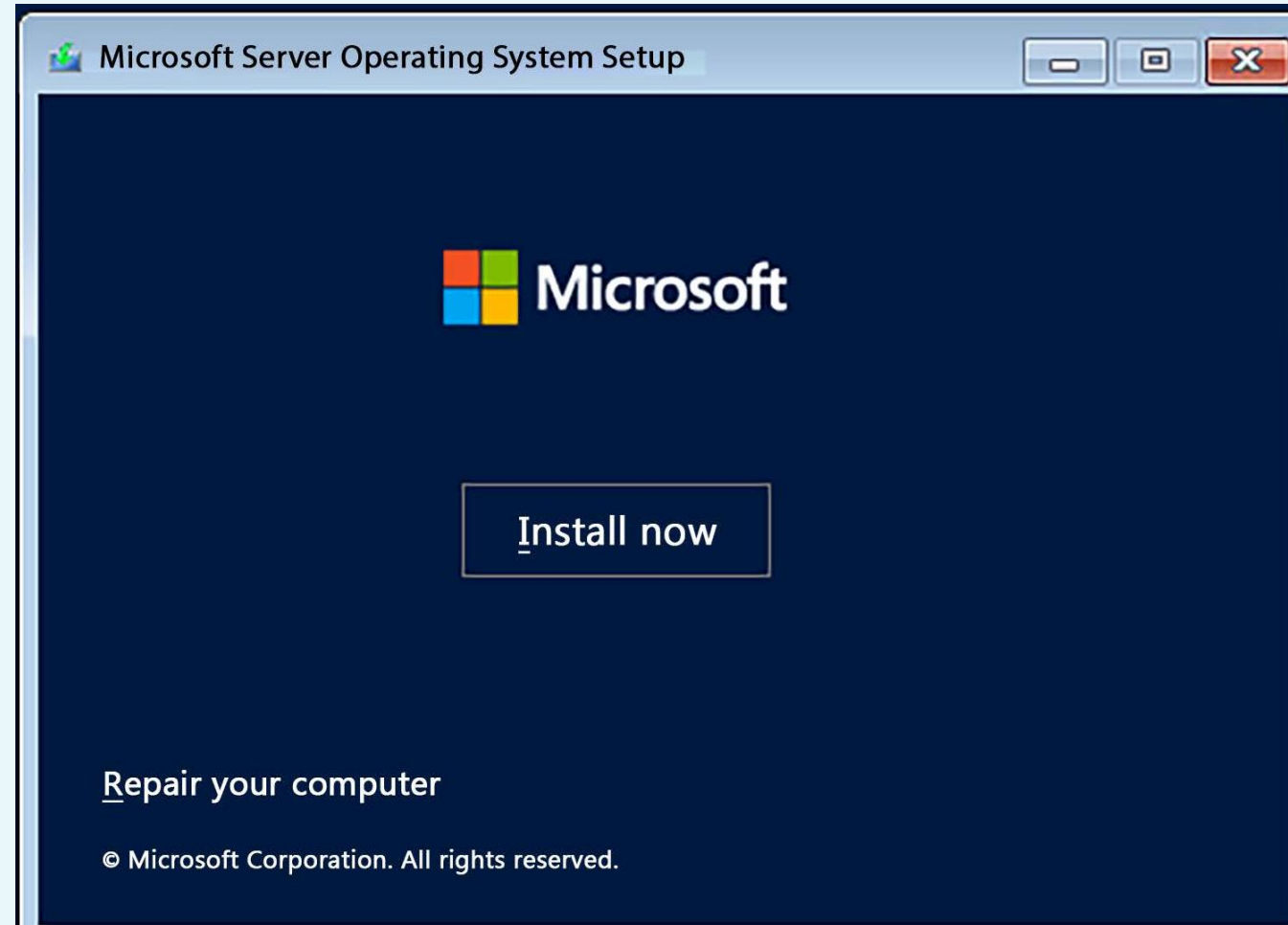
- CPU: 1.4 GHz 64-bit that supports: NX, DEP, CMPXCHG16b, LAHF/SAHF, PrefetchW, and SLAT.
- RAM: 512 MB ECC memory minimum for Server Core, or a recommended 2 GB minimum for a server running Desktop Experience.
- Disk: 32 GB
 - Desktop Experience consumes about 4 GB more space than Server Core
 - Requires a PCI Express (PCIe) storage adapter
 - ATA/PATA/IDE are not allowed for boot drives
- Windows Server 2022 supports 48 TB of memory and 2,048 logical cores across 64 physical sockets



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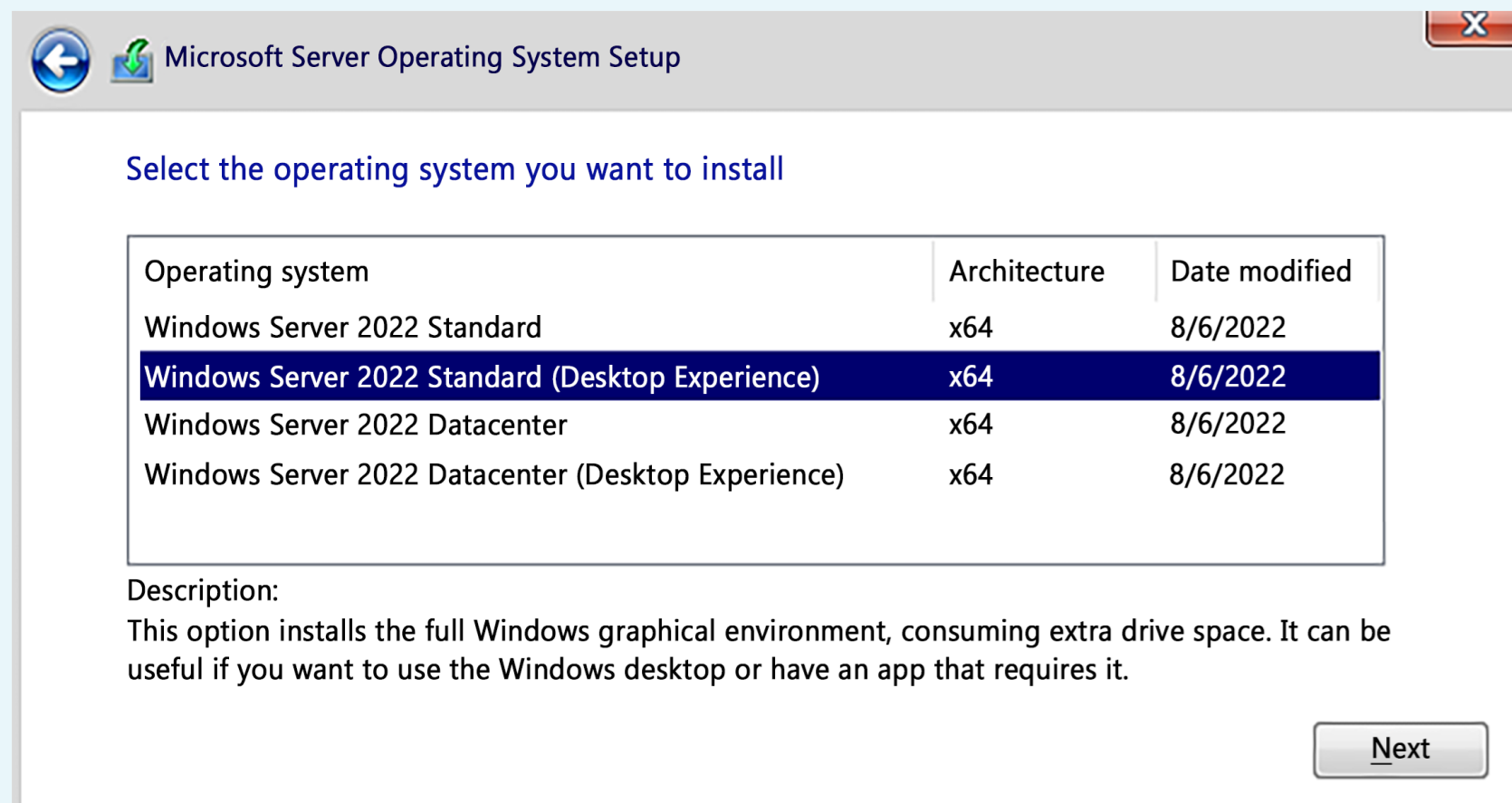
Installing Windows Server 2022

- Burning that ISO: Download a .ISO file from Microsoft, burn that .ISO to a DVD or Creating a bootable USB stick
- Running the installer
 - plug newly created DVD or bootable USB into the new server hardware.
 - Boot to it, and the installation wizard for Windows Server 2022 appear
 - After choosing installation language, the next screen is Install now



Installing Windows Server 2022

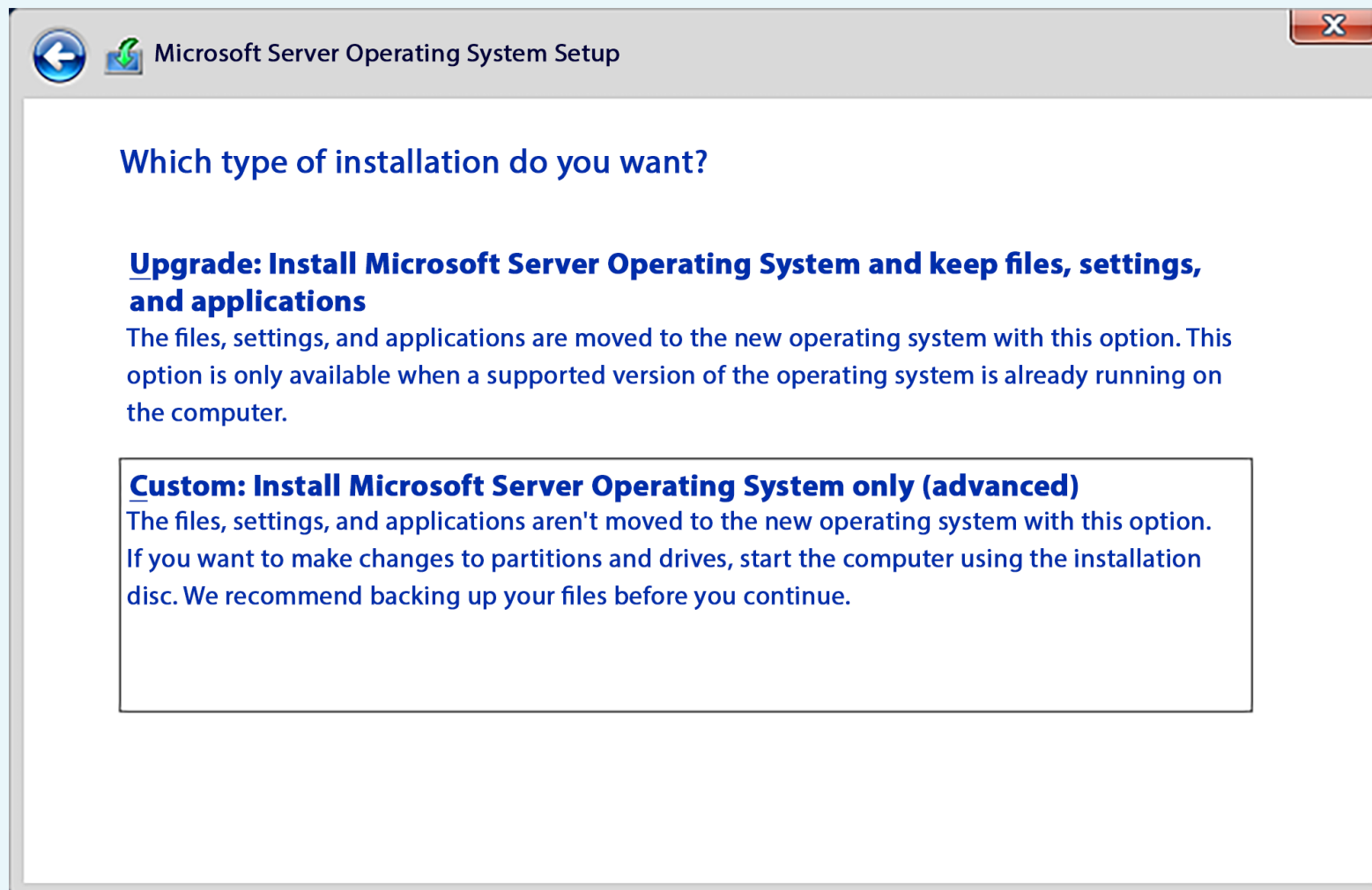
- To run in trial mode click on “I don’t have a product key”
- Select installation options for Windows Server 2022





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Installing Windows Server 2022





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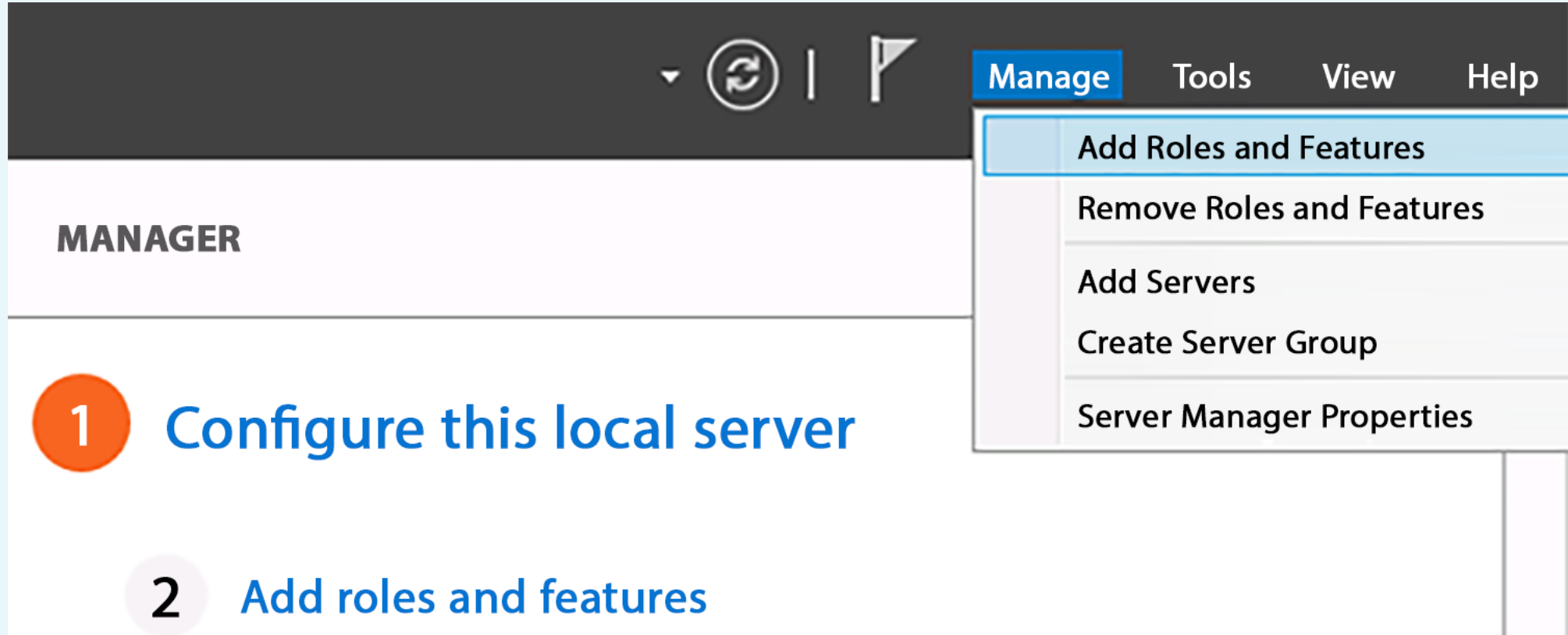
Installing roles and features

- Installing the operating system gets your foot in the door
- Not actually do anything useful with your server at this point
- Server's job is to serve up data
- Need to utilize roles and features

Installing roles and features

Installing a role using the wizard

- Server Manager launches automatically every time you log in to Windows Server 2022



Installing roles and features

Installing a role using the wizard

- Select Role-based or feature-based installation

☒ **Role-based or feature-based installation**

Configure a single server by adding roles, role services, and features.

☐ **Remote Desktop Services installation**

Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment



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Installing roles and features

Installing a role using the wizard

Select destination server

DESTINATION SERVER
DC1

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

- ☒ Select a server from the server pool
- ☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
DC1	10.10.10.10	Microsoft Windows Server 2022 Standard

Installing roles and features

Installing a role using the wizard

- Now we have our list of roles available to be installed
- In this example, we will the core infrastructural roles:
 - Active Directory Domain Services role,
 - DNS server and
 - DHCP server.

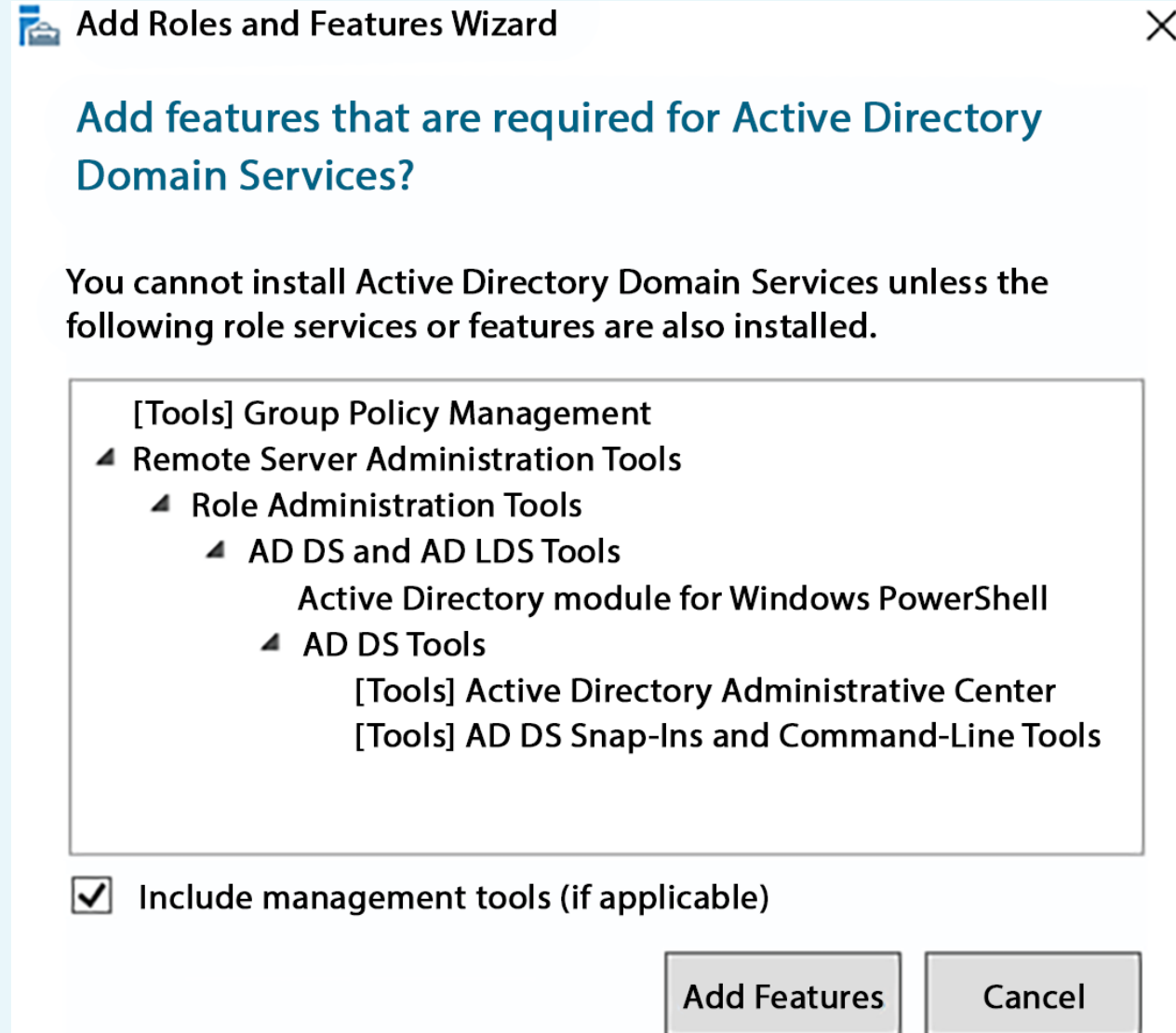
Roles

- ☐ Active Directory Certificate Services
- ☒ Active Directory Domain Services
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☒ DHCP Server
- ☒ DNS Server
- ☐ Fax Server
- ▶ ☒ File and Storage Services (1 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☐ Web Server (IIS)
- ☐ Windows Deployment Services
- ☐ Windows Server Update Services

Installing roles and features

Installing a role using the wizard

- When clicked on Active Directory Domain Services role, a pop-up message that requires some additional features in order to work properly.
- This is normal behavior
- All you need to do is click on the Add Features button



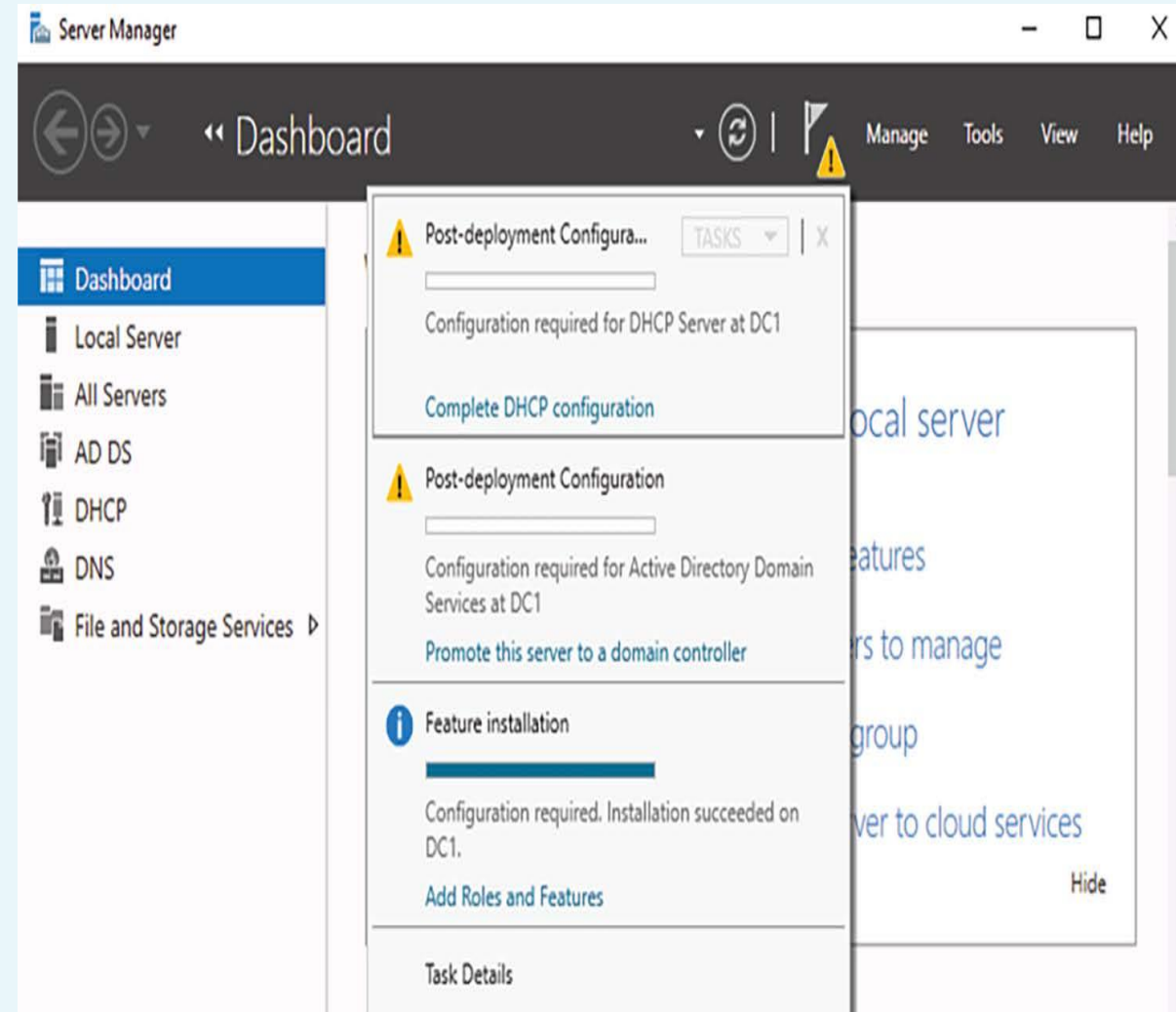


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Installing roles and features

Installing a role using the wizard

- After the installation process completed, you may or may not be prompted to restart the server
- Once back inside Server Manager, there may be a yellow exclamation mark near the top
- Clicking here displays messages about further configurations required to complete the setup of new roles and finalize their use on the server





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Installing roles and features

Installing a feature using PowerShell

- Get-WindowsFeature

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWin
dows

PS C:\Users\Administrator> Get-WindowsFeature

Display Name                                Name                                Install State
-----
[ ] Active Directory Certificate Services    AD-Certificate                      Available
[ ] Certification Authority                  ADCS-Cert-Authority                 Available
[ ] Certificate Enrollment Policy Web Service ADCS-Enroll-Web-Pol                 Available
[ ] Certificate Enrollment Web Service       ADCS-Enroll-Web-Svc                 Available
[ ] Certification Authority Web Enrollment   ADCS-Web-Enrollment                 Available
[ ] Network Device Enrollment Service        ADCS-Device-Enrollment              Available
[ ] Online Responder                        ADCS-Online-Cert                     Available
[X] Active Directory Domain Services         AD-Domain-Services                  Installed
[ ] Active Directory Federation Services     ADFS-Federation                     Available
[ ] Active Directory Lightweight Directory Services ADLDS                               Available
[ ] Active Directory Rights Management Services ADRMS                               Available
[ ] Active Directory Rights Management Server ADRMS-Server                         Available
[ ] Identity Federation Support              ADRMS-Identity                       Available
[ ] Device Health Attestation                DeviceHealthAttestat...              Available
[X] DHCP Server                             DHCP                                 Installed
[X] DNS Server                              DNS                                 Installed
[ ] Fax Server                              Fax                                 Available
[X] File and Storage Services                FileAndStorage-Services              Installed
[ ] File and iSCSI Services                  File-Services                        Available
[ ] File Server                             FS-FileServer                       Available
[ ] BranchCache for Network Files            FS-BranchCache                       Available
```

Installing roles and features

Installing a feature using PowerShell

- To see only the features that begin with the letters TEL

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-WindowsFeature -Name TEL*

Display Name          Name          Install State
-----
[ ] Telnet Client     Telnet-Client Available
```

Installing roles and features

Installing a feature using PowerShell

- To quickly show only the roles and features currently installed on a server

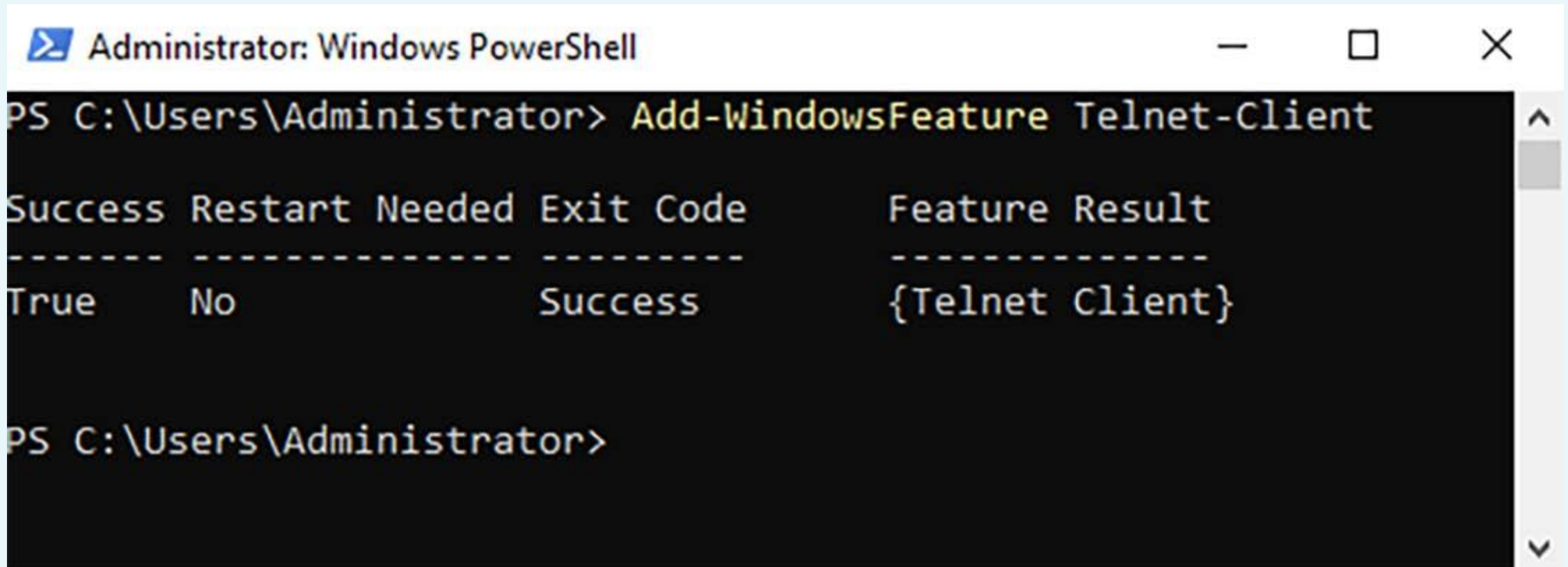
```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Get-WindowsFeature | Where Installed
```

Display Name	Name	Install State
[X] Active Directory Domain Services	AD-Domain-Services	Installed
[X] DHCP Server	DHCP	Installed
[X] DNS Server	DNS	Installed
[X] File and Storage Services	FileAndStorage-Services	Installed
[X] Storage Services	Storage-Services	Installed
[X] .NET Framework 4.8 Features	NET-Framework-45-Fea...	Installed
[X] .NET Framework 4.8	NET-Framework-45-Core	Installed
[X] WCF Services	NET-WCF-Services45	Installed
[X] TCP Port Sharing	NET-WCF-TCP-PortShar...	Installed
[X] Group Policy Management	GPMC	Installed
[X] Microsoft Defender Antivirus	Windows-Defender	Installed
[X] Remote Server Administration Tools	RSAT	Installed
[X] Role Administration Tools	RSAT-Role-Tools	Installed
[X] AD DS and AD LDS Tools	RSAT-AD-Tools	Installed
[X] Active Directory module for Windows ...	RSAT-AD-PowerShell	Installed
[X] AD DS Tools	RSAT-ADDS	Installed
[X] Active Directory Administrative ...	RSAT-AD-AdminCenter	Installed
[X] AD DS Snap-Ins and Command-Line ...	RSAT-ADDS-Tools	Installed
[X] DHCP Server Tools	RSAT-DHCP	Installed
[X] DNS Server Tools	RSAT-DNS-Server	Installed
[X] System Data Archiver	System-DataArchiver	Installed
[X] Telnet Client	Telnet-Client	Installed
[X] Windows PowerShell	PowerShellRoot	Installed
[X] Windows PowerShell 5.1	PowerShell	Installed
[X] WoW64 Support	WoW64-Support	Installed
[X] XPS Viewer	XPS-Viewer	Installed

Installing roles and features

Installing a feature using PowerShell

- To install Telnet-Client



```
Administrator: Windows PowerShell

PS C:\Users\Administrator> Add-WindowsFeature Telnet-Client

Success Restart Needed Exit Code      Feature Result
-----
True      No              Success      {Telnet Client}
```

PS C:\Users\Administrator>

Centralized management and monitoring

- To manage a server => log directly into the server that you need to work on
 - ✓ walking up to the server and logging on with the keyboard and mouse plugged right into that hardware
 - ✓ Logging into our servers remotely using RDP
- Need some tools that we can utilize to make our management and monitoring, and even configuration processes, more efficient



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Centralized management and monitoring

Server Manager

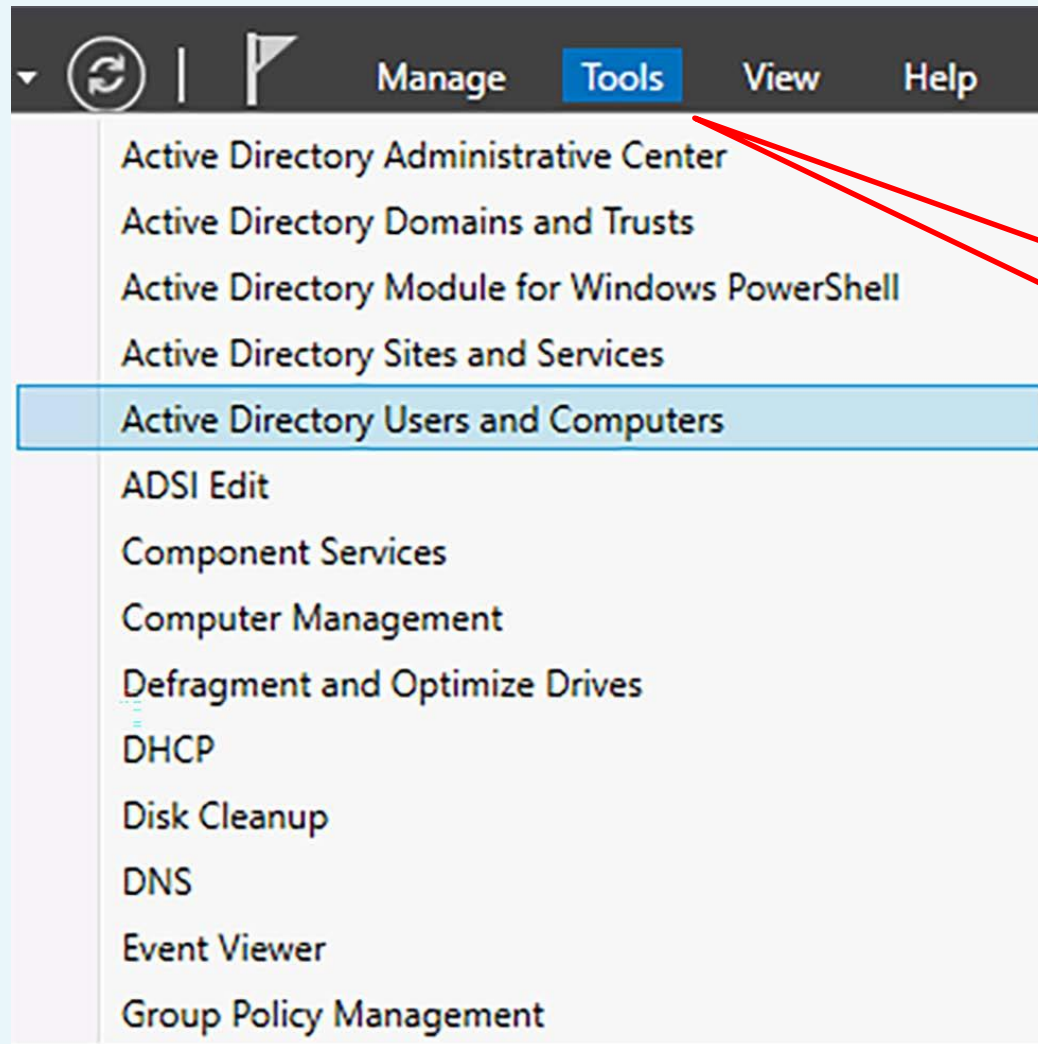
- Auto-launching program when logging in to any of servers
- Help to manage the server
- Give a quick look into what is currently installed on the server

a very quick view of whether or not the services running on this server functioning properly

The screenshot shows the Windows Server Manager Dashboard. The left-hand navigation pane is highlighted with a red box and contains the following items: Dashboard (selected), Local Server, All Servers, AD DS, DHCP, DNS, and File and Storage Services. The main area of the dashboard is titled 'WELCOME TO SERVER MANAGER' and features a 'QUICK START' section with a numbered list of five steps: 1. Configure this local server, 2. Add roles and features, 3. Add other servers to manage, 4. Create a server group, and 5. Connect this server to cloud services. Below this, there is a 'WHAT'S NEW' section and a 'LEARN MORE' link. At the bottom of the dashboard, a section titled 'ROLES AND SERVER GROUPS' is highlighted with a red box. It shows 'Roles: 4 | Server groups: 1 | Servers total: 1'. Under this, there are two columns. The left column is for 'AD DS' and shows a green status icon, 'Manageability', 'Events', 'Services', 'Performance', and 'BPA results'. The right column is for 'DHCP' and shows a green status icon, 'Manageability', 'Events', 'Services', 'Performance', and 'BPA results'.

Centralized management and monitoring

Server Manager

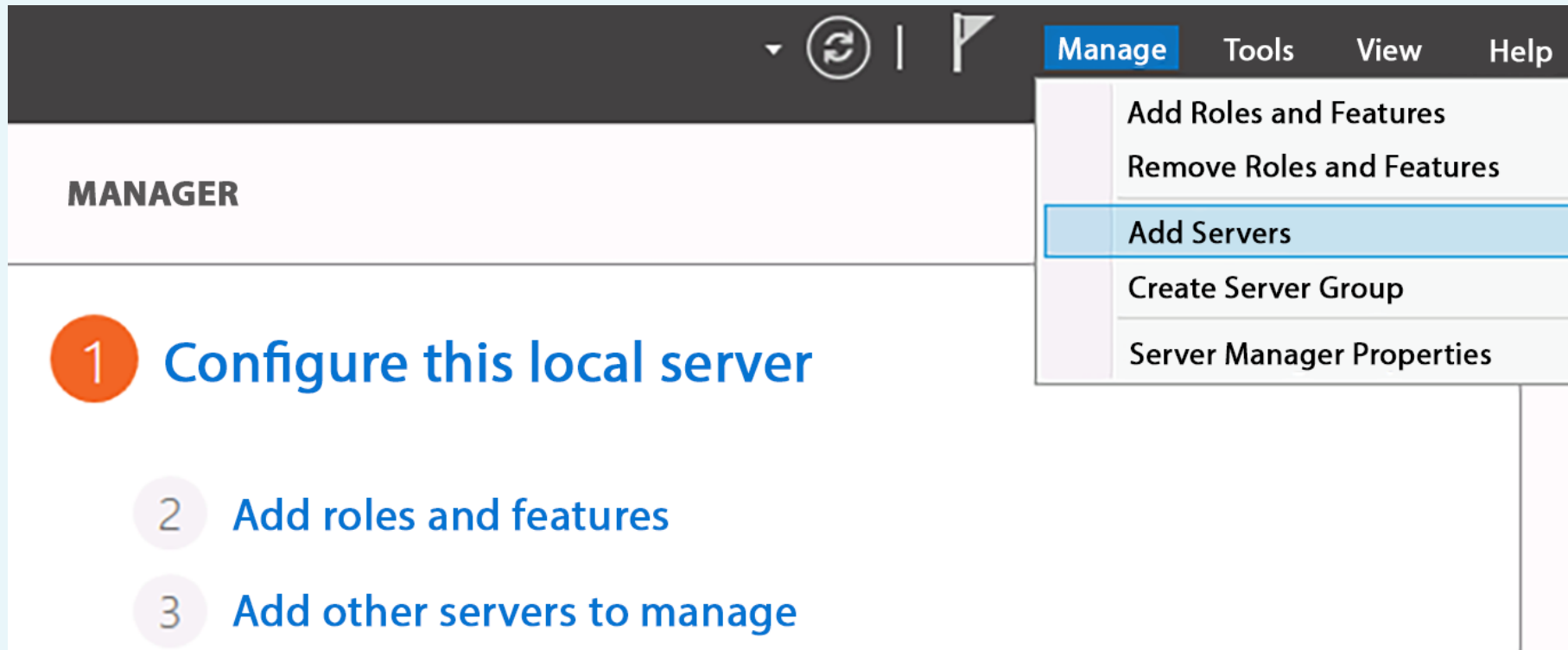


a list of all the available
Administrative Tools

Centralized management and monitoring

Server Manager

- Managing many servers from a single server



The screenshot displays the Windows Server Manager interface. The top navigation bar includes icons for refresh, help, and a flag, followed by the menu items: **Manage**, Tools, View, and Help. The main area is titled **MANAGER** and contains a list of tasks:

- 1 **Configure this local server**
- 2 Add roles and features
- 3 Add other servers to manage

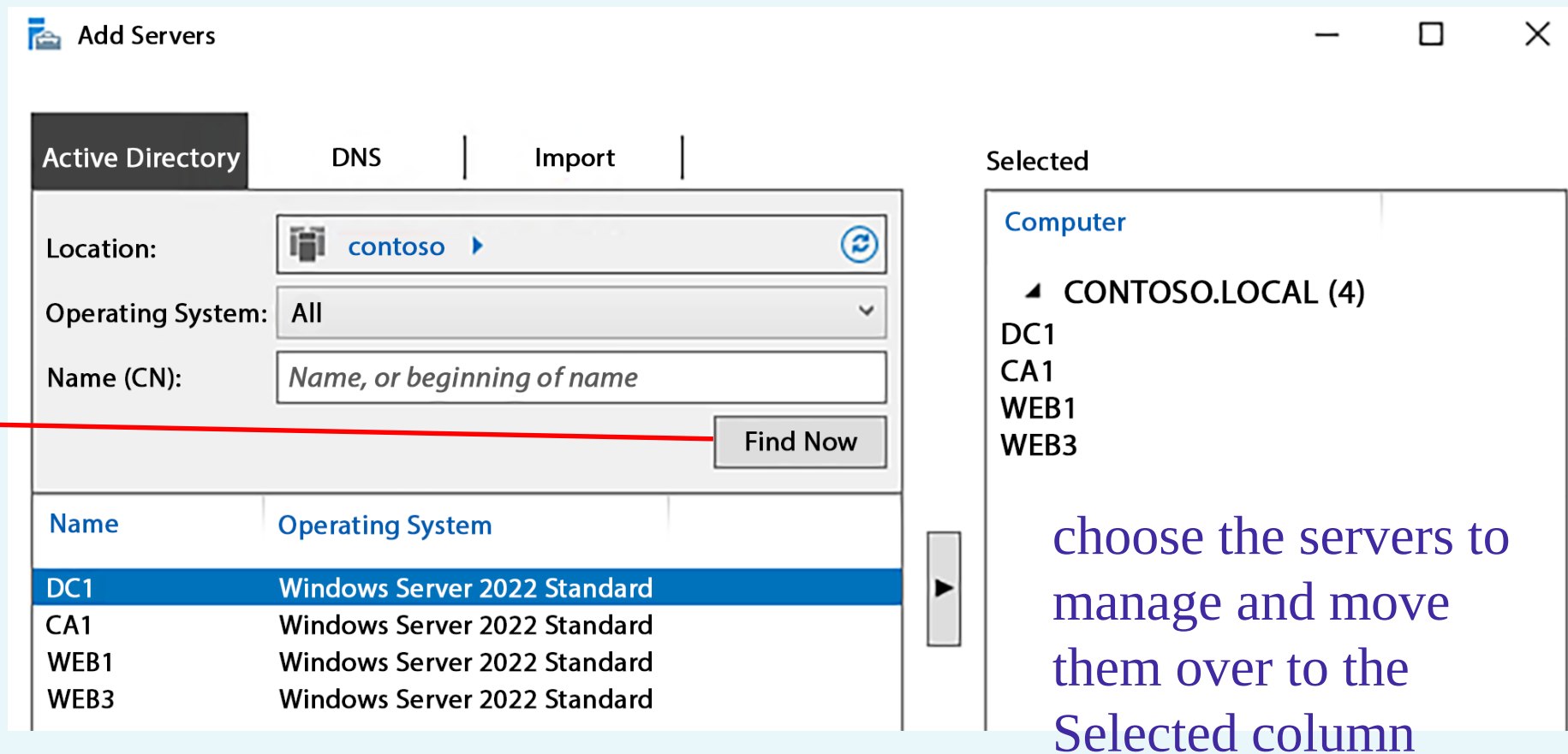
The **Manage** menu is open, showing the following options:

- Add Roles and Features
- Remove Roles and Features
- Add Servers** (highlighted)
- Create Server Group
- Server Manager Properties

Centralized management and monitoring

Server Manager

- Managing many servers from a single server



Find machines available within the network

Find Now

Name	Operating System
DC1	Windows Server 2022 Standard
CA1	Windows Server 2022 Standard
WEB1	Windows Server 2022 Standard
WEB3	Windows Server 2022 Standard

Selected

Computer

CONTOSO.LOCAL (4)

DC1
CA1
WEB1
WEB3

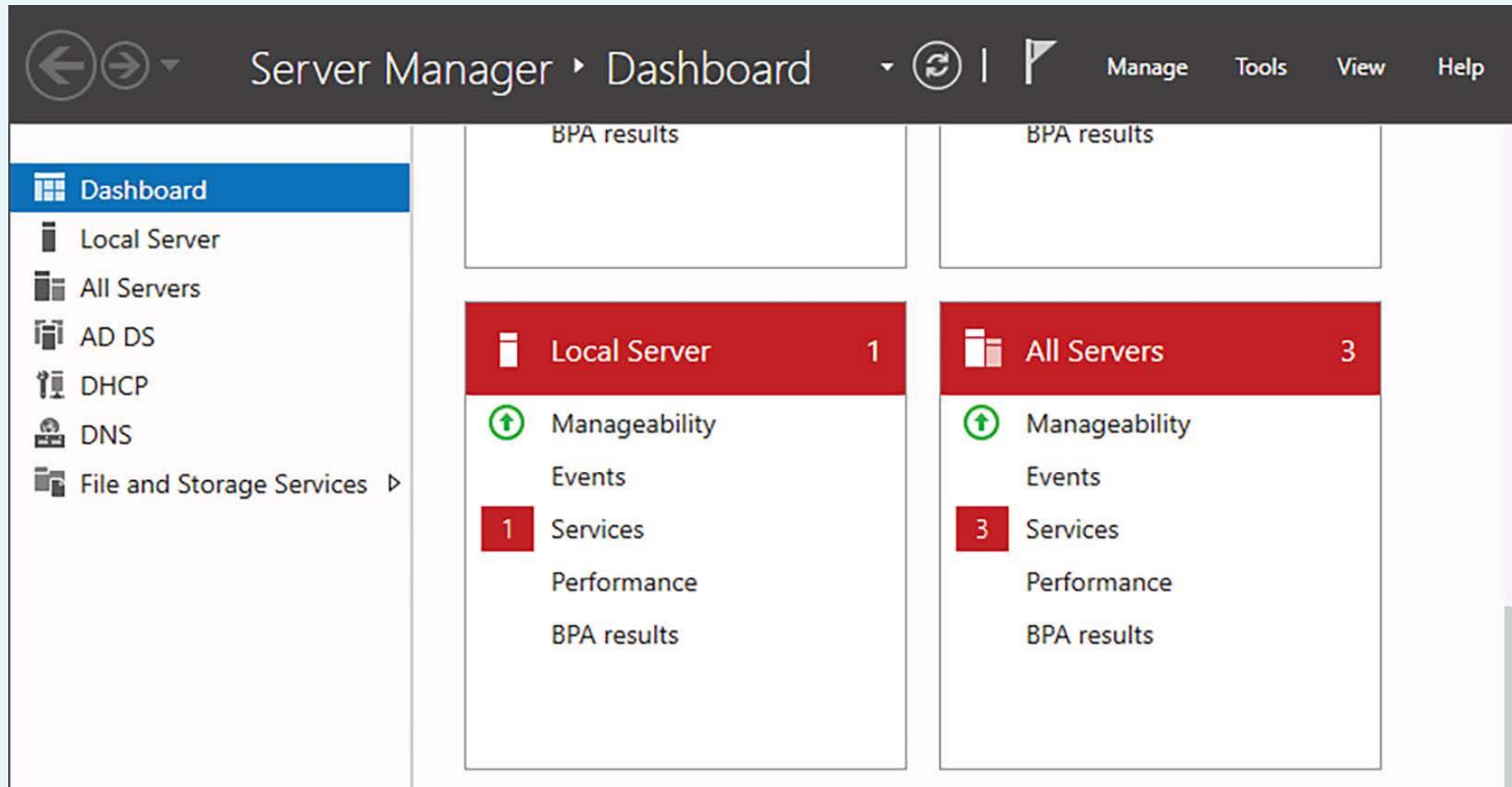
choose the servers to manage and move them over to the Selected column



Centralized management and monitoring

Server Manager

- Managing many servers from a single server



Centralized management and monitoring

Remote Server Administration Tools (RSAT)

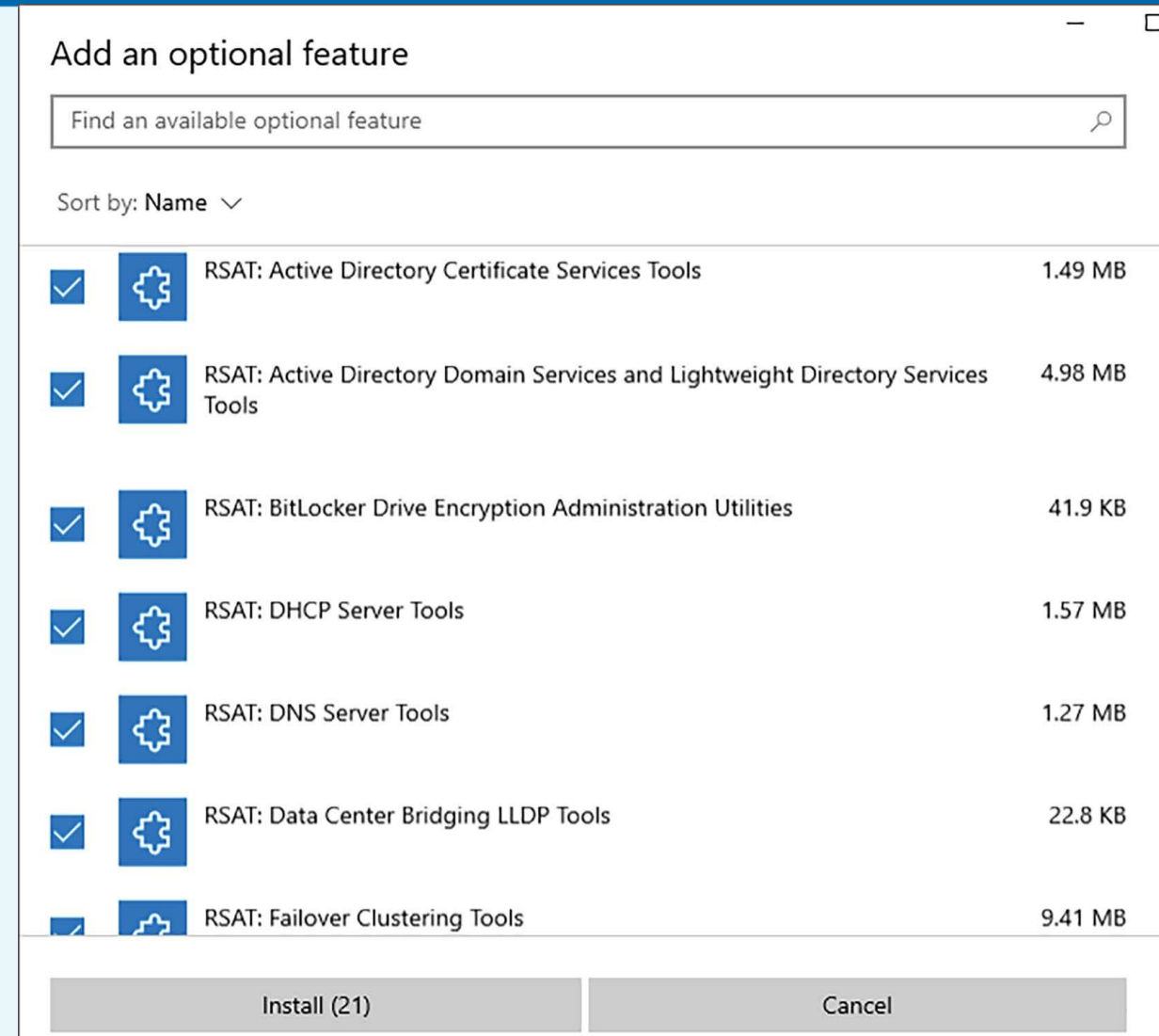
- To manage and monitor all of the servers from client computer
- a regular Windows 10 client computer online and running in our network, also domain-joined
- Add an optional feature to this Windows 10 computer to give it the RSAT toolset.

Centralized management and monitoring

Remote Server Administration Tools (RSAT)

- 1) Open up **Settings** and type the word **optional** into the search bar.
- 2) Click on Manage optional features
- 3) Click the button to Add a feature
- 4) check the box next to every item that begins with RSAT
- 5) Click Install

Windows 10 older than 1809: Download and install the whole RSAT package from <https://www.microsoft.com/en-us/download/details.aspx?id=45520>





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Centralized management and monitoring

Remote Server Administration Tools (RSAT)

- No program called Remote Server Administration Tool.
- In Start menu, you will now find Server Manager
- Open up Server Manager on your desktop computer
- Walk through the step to Add other servers to manage

Centralized management and monitoring

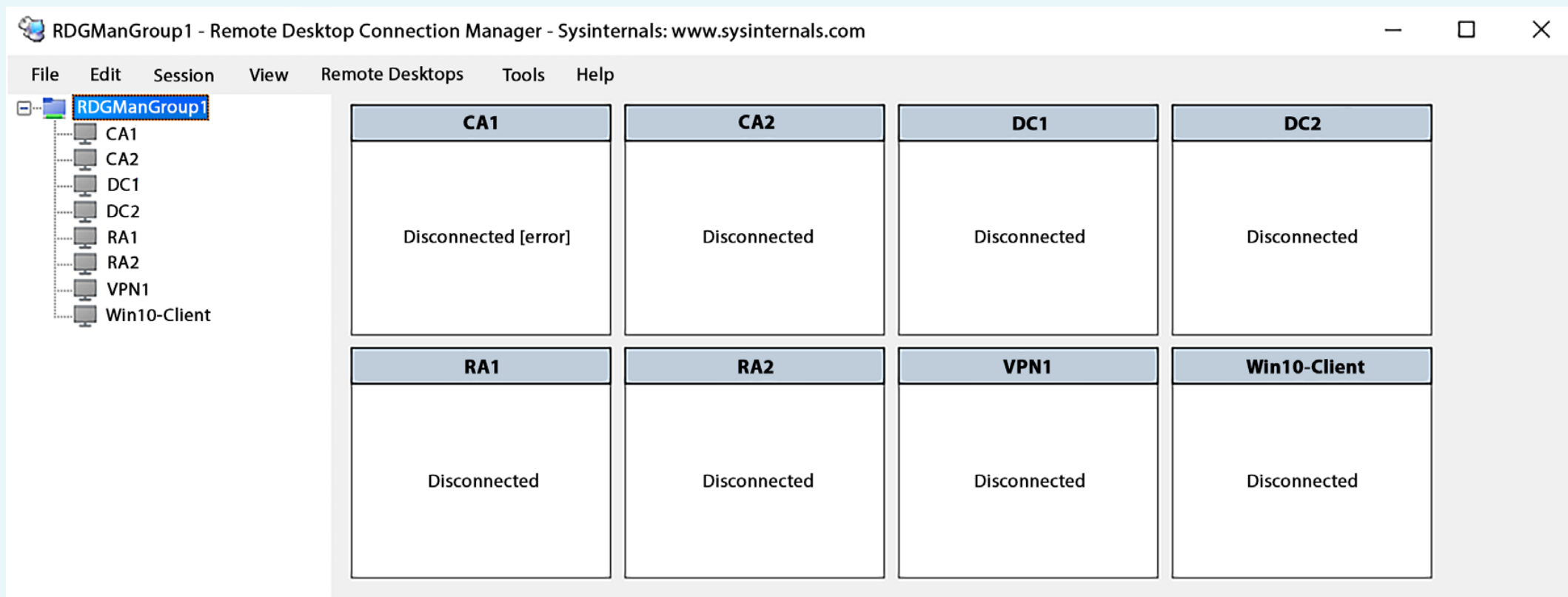
Does this mean RDP is dead?

- Certainly not!
- Need to access our servers directly sometimes
- Many administrators out there will continue using RDP and full desktop-based access for all management and monitoring of their servers simply because that is what they are more comfortable with

Centralized management and monitoring

Remote Desktop Connection Manager

- Useful platform for storing all of the different RDP connections



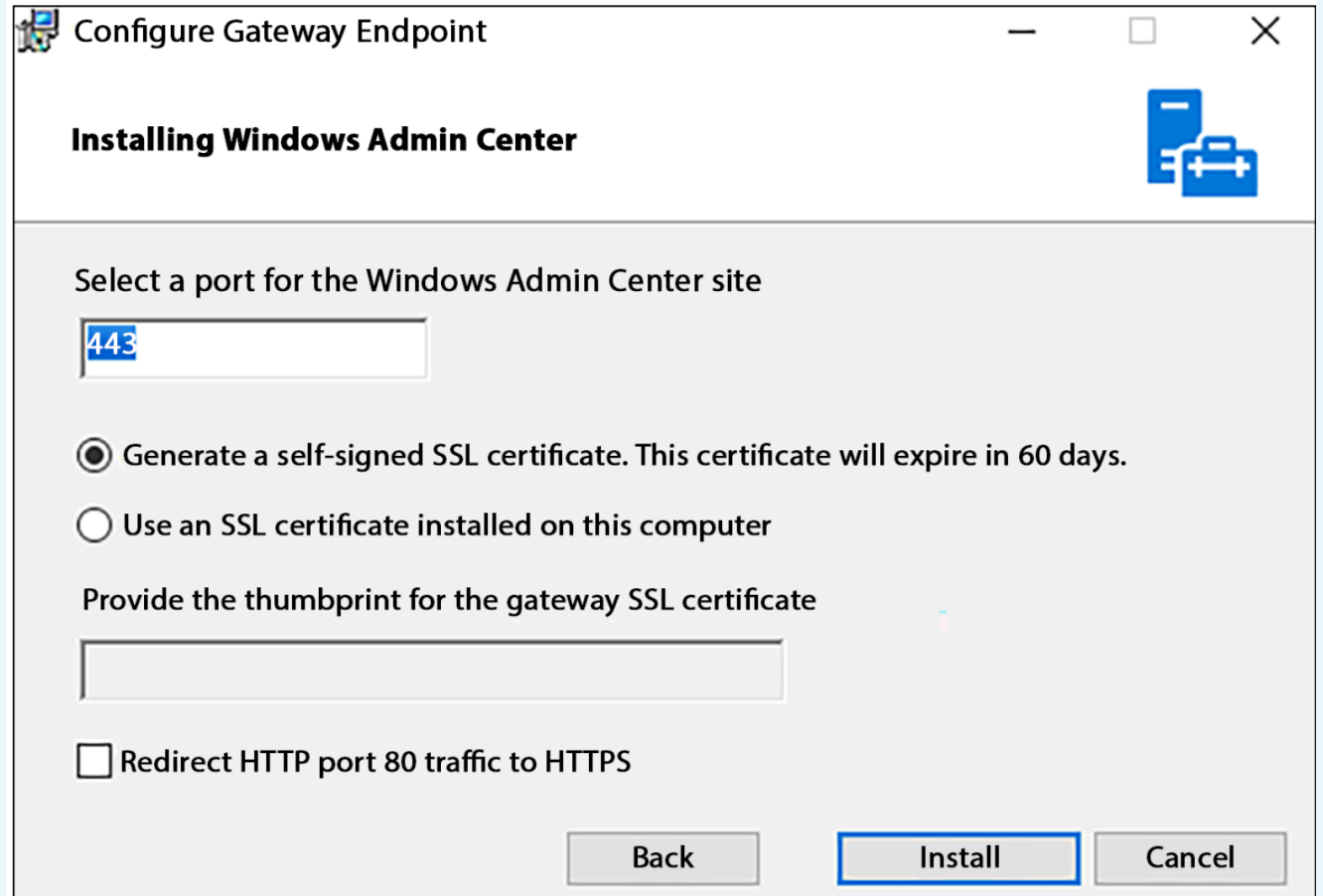
Windows Admin Center (WAC)

- A server and client management platform designed to administer machines in a more efficient manner.
- A browser-based tool: once installed, you access WAC from a web browser
- No need to install a management tool or application onto your workstation
- To manage the servers (all the way back to Server 2008 R2) and the server clusters
- Ability to manage servers hosted on-premises as well as inside Azure

Windows Admin Center (WAC)

Installing Windows Admin Center

- Log into a server in the network
- Download WAC here:
<https://aka.ms/WACDownload>
- Run the installer
- Windows Admin Center cannot be installed Active Directory Domain Services (AD DS)



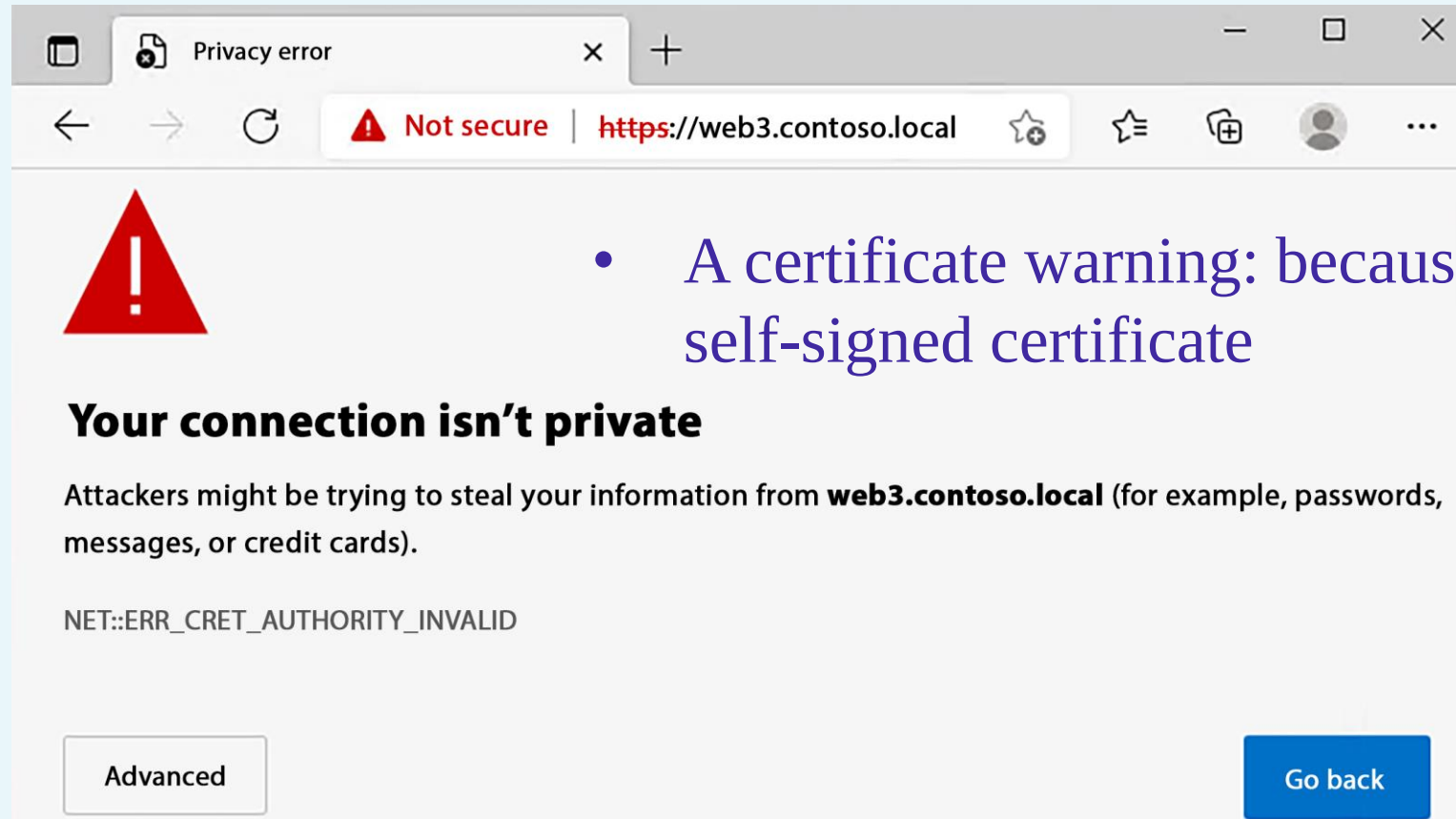
The screenshot shows the 'Configure Gateway Endpoint' window titled 'Installing Windows Admin Center'. It contains the following elements:

- A header bar with the title 'Installing Windows Admin Center' and a blue icon of a server and a plus sign.
- A section titled 'Select a port for the Windows Admin Center site' with a text input field containing '443'.
- Two radio button options:
 - ☒ Generate a self-signed SSL certificate. This certificate will expire in 60 days.
 - ☐ Use an SSL certificate installed on this computer
- A section titled 'Provide the thumbprint for the gateway SSL certificate' with an empty text input field.
- A checkbox labeled 'Redirect HTTP port 80 traffic to HTTPS' which is currently unchecked.
- Three buttons at the bottom: 'Back', 'Install' (highlighted with a blue border), and 'Cancel'.

Windows Admin Center

Launching Windows Admin Center

- Open up a supported browser from any machine and browse to the WAC URL

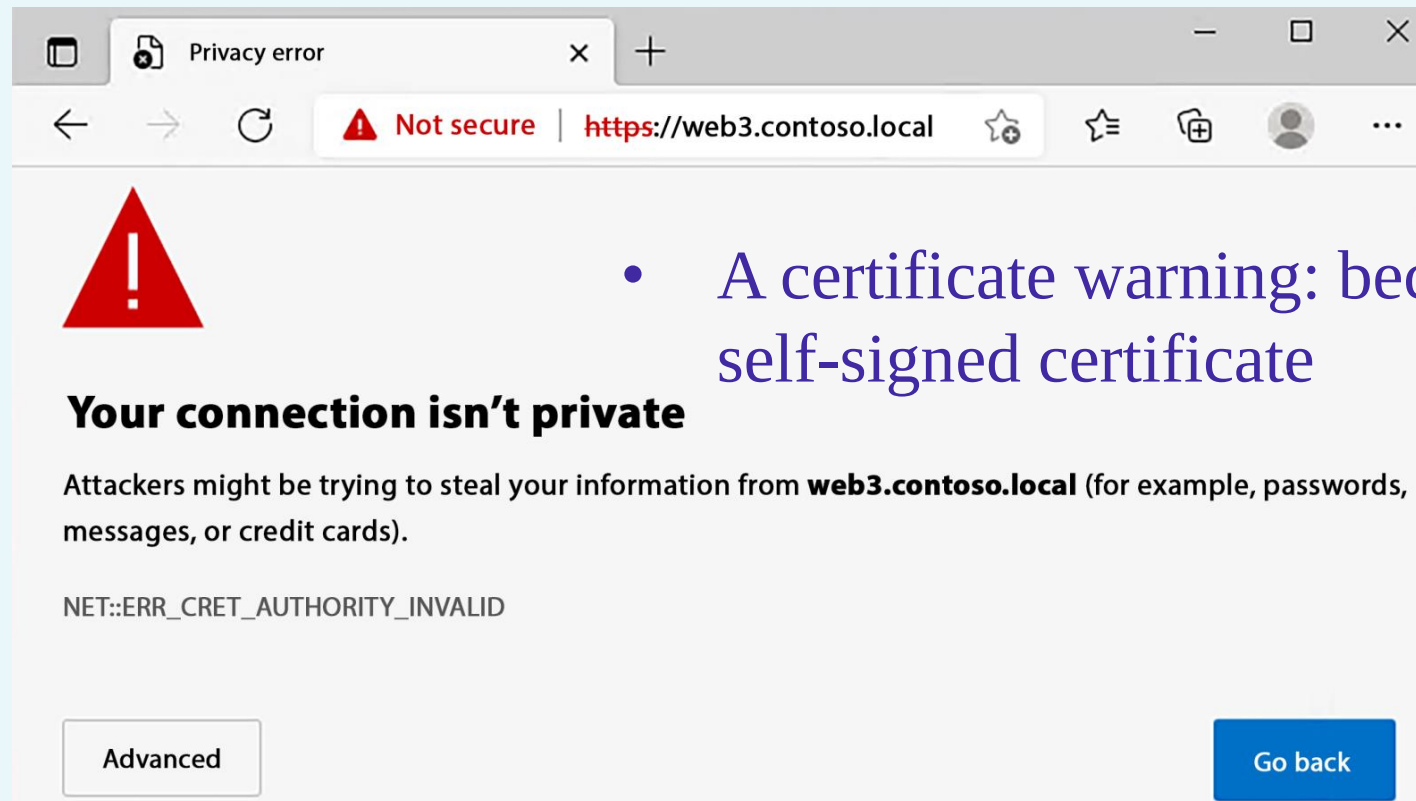


- A certificate warning: because I of using a self-signed certificate

Windows Admin Center

Launching Windows Admin Center

- Open up a supported browser from any machine and browse to the WAC URL



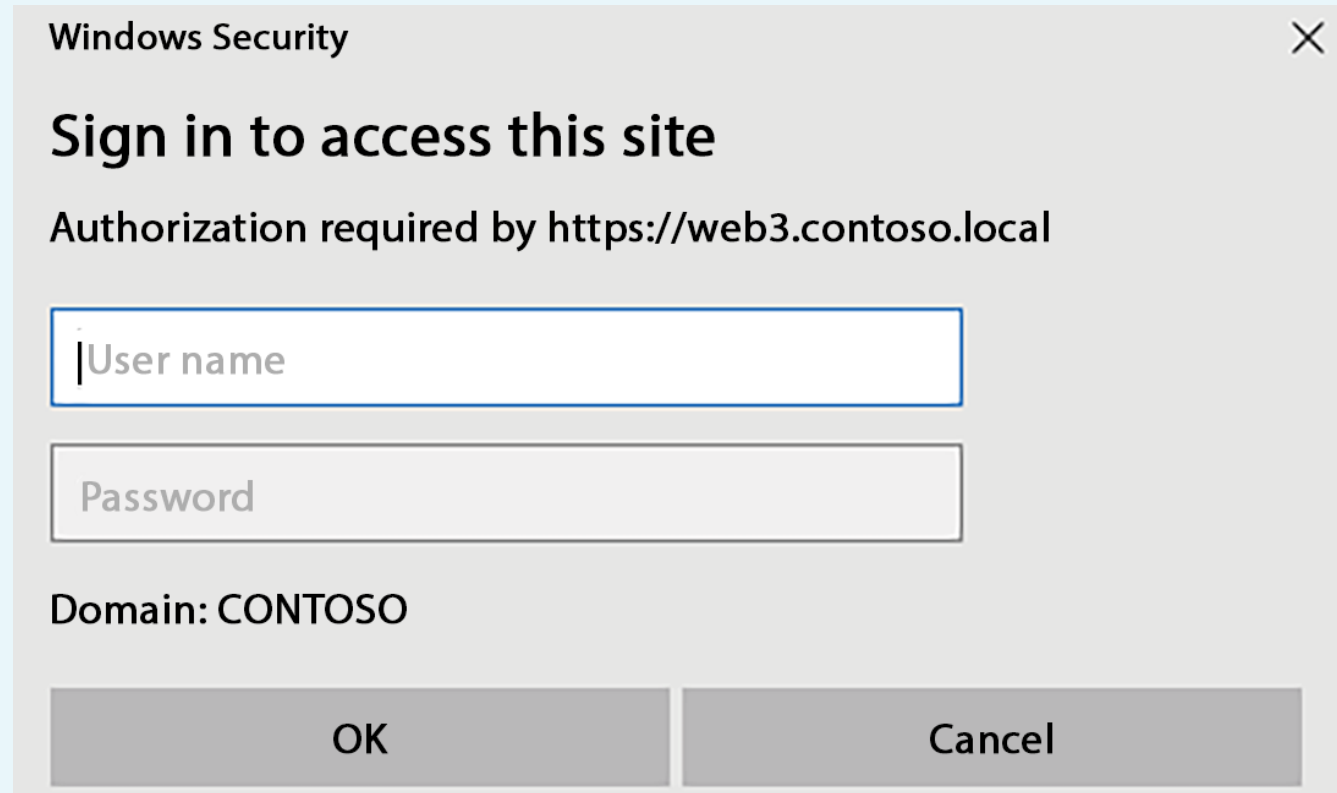
- A certificate warning: because I of using a self-signed certificate

Windows Admin Center

Launching Windows Admin Center

- Even though logged into a domain-joined Windows 10 computer, the WAC website does not automatically try to inject those credentials for its own use but rather pauses to ask who you are

Input the domain
credentials



Windows Security

Sign in to access this site

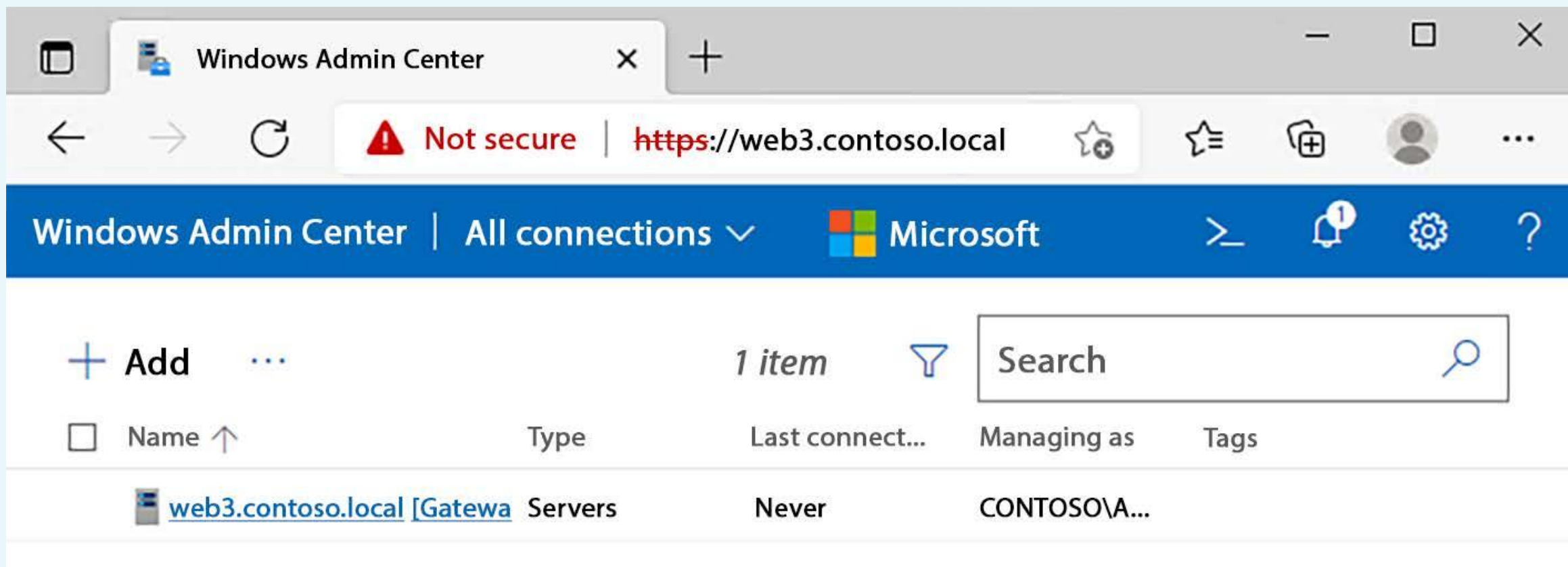
Authorization required by <https://web3.contoso.local>

Domain: CONTOSO

Windows Admin Center

Launching Windows Admin Center

- WAC interface



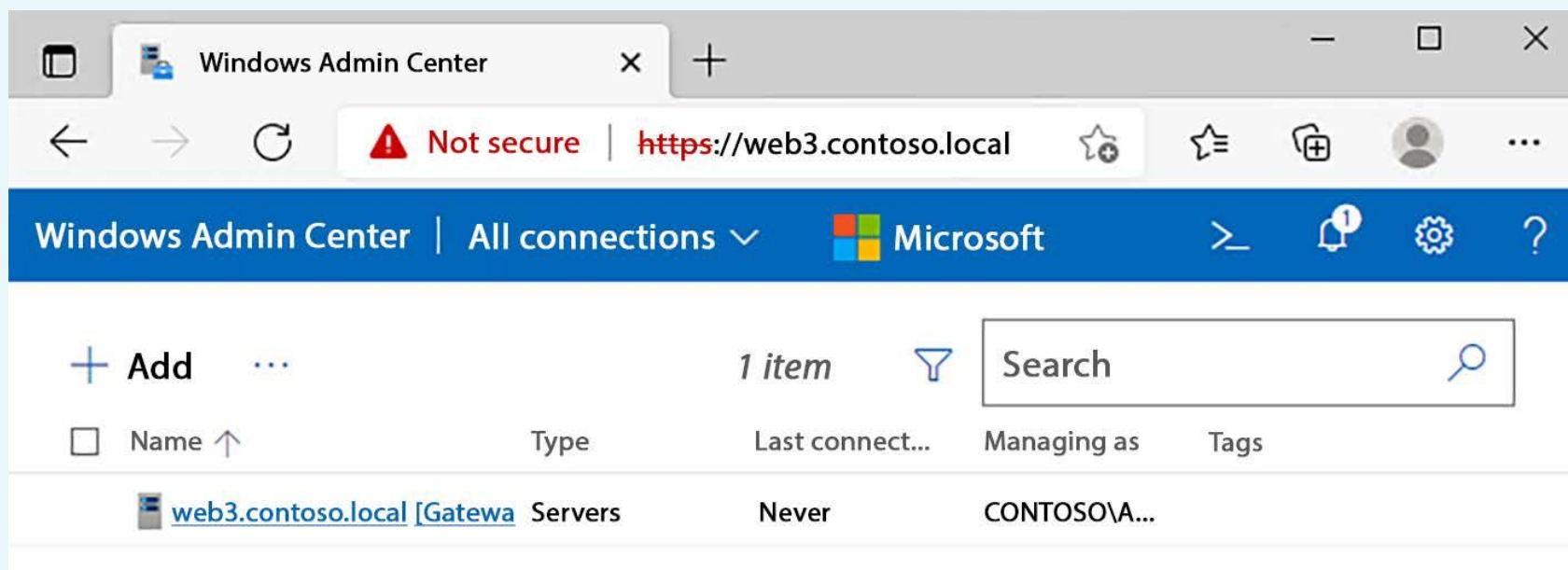
The screenshot shows the Windows Admin Center web interface. The browser tab is titled "Windows Admin Center" and the address bar shows "https://web3.contoso.local" with a "Not secure" warning. The interface has a blue header bar with "Windows Admin Center | All connections" and a Microsoft logo. Below the header, there is a section with "Add" and "1 item". A search bar is present. The main content area displays a table with one item:

☐	Name ↑	Type	Last connect...	Managing as	Tags
☐	web3.contoso.local [Gatewa	Servers	Never	CONTOSO\A...	

Windows Admin Center

Adding more servers to WAC

- Click the + Add button that is shown onscreen



- You will be presented with choices to add a new server, a new PC, a Windows Server failover cluster, Azure Stack HCI, or even an Azure VM.

Windows Admin Center

Adding more servers to WAC

- The option to **Search Active Directory** to search for a full list of machines in the domain
- select the checkboxes next to each server that you would like to administer via Windows Admin Center and click the **Add** button

Connection tags ⓘ

[+ Add tags](#)

[Add one](#) [Import a list](#) [Search Active Directory](#)

[Search](#)

4 items

☐ Name ↑	Type	Login name
CA1.contoso.local	computer	contoso.local\CA1\$
DC1.contoso.local	computer	contoso.local\DC1\$
WEB1.contoso.local	computer	contoso.local\WEB1\$
WEB3.contoso.local	computer	contoso.local\WEB3\$

[Add](#) [Cancel](#)

Windows Admin Center

Adding more servers to WAC

- Server information in WAC

Windows Admin Center All connections ▾			Microsoft			
+ Add  Connect  Manage as ...			4 items		Search 	
☐	Name ↑	Type	Last connected	Managing as	Tags	
	ca1.contoso.local	Servers	Never	CONTOSO\Administrat...		
	dc1.contoso.local	Servers	Never	CONTOSO\Administrat...		
	web1.contoso.local	Servers	Never	CONTOSO\Administrat...		
	web3.contoso.local [Gateway.]	Servers	Never	CONTOSO\Administrat...		



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Windows Admin Center

Managing a server with WAC

- Clicking on the server name
- The ability to run backups on server, view and install certificates
- Monitor the performance of the server
- Launch a remote PowerShell connection to the server
- One-stop shop for remotely managing your servers

The screenshot displays the Windows Admin Center interface for a server named **dc1.contoso.local**. The top navigation bar includes "Windows Admin Center | Server Manager" and the Microsoft logo. A left-hand sidebar lists various management tools, with "Overview" selected. The main content area provides a comprehensive overview of the server's hardware and software configuration. Key details include the operating system version (10.0.20348), 4 GB of installed memory, and 115.69 GB of free disk space out of 126.33 GB. It also lists the processor as an Intel(R) Xeon(R) CPU E5-2430L 0 @ 2.00GHz, identifies the manufacturer as Microsoft Corporation, and confirms the server is a Virtual Machine. Security status is shown with Microsoft Defender Antivirus real-time protection turned on. Performance metrics for the CPU are displayed, showing 9.01% utilization and 46389 handles. A line graph at the bottom tracks CPU utilization from 60 seconds ago to the present. Other visible metrics include 1 logical processor, 1 NIC, and 1 logged-in user.

Server: dc1.contoso.local		
Version	10.0.20348	Installed memory (RAM)
		4 GB
		Disk space (Free / Total)
		115.69 GB / 126.33 GB
Processors	Intel(R) Xeon(R) CPU E5-2430L 0 @ 2.00GHz	Manufacturer
		Microsoft Corporation
		Model
		Virtual Machine
Logical processors	1	Microsoft Defender Antivirus
		Real-time protection: On
		NIC(s)
		1
Azure Backup status	Not protected	Up time
		0:11:44:47
		Logged in users
		1
CPU		
Utilization		Handles
9.01%		46389
Speed		
2GHz		
Processes		
109		
Threads		



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Windows Admin Center

Changes are easy as pie

- Capabilities to manipulate servers as well, straight from this web interface
- Can add roles or features to servers, create scheduled tasks, start or stop services, edit the registry and add Windows Firewall rules ...
- **Example:** Enable RDP from inside WAC

The screenshot shows the Windows Admin Center web interface. The top navigation bar is blue with "Windows Admin Center | Server Manager" and the Microsoft logo. The main content area is for "dc1.contoso.local". On the left, under "Tools", there is a list of tools: Networks, Packet monitoring, Performance Monitor, PowerShell, Processes, Registry, Remote Desktop (highlighted with a red circle), Roles & features, Scheduled tasks, and Security. On the right, under "Settings", the "General" tab is selected, showing options like File shares (SMB server), Environment variables, Azure Arc for servers, Power configuration, Remote Desktop (selected), and Role-based Access Control. Below this, the "Remote Desktop" section is expanded, showing two radio button options: "Don't allow remote connections to this computer" (unselected) and "Allow remote connections to this computer" (selected). Under the selected option, there is a checked checkbox for "Allow connections only from computers running Remote Desktop with Network Level Authentication (recommended)".



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Windows Admin Center

Azure integrations

- WAC that there are numerous tools related to Azure
- Can be used to administer both on-premises servers as well as Azure servers
- Used to bind your on-premises environment together with your Azure environment through things like Azure File Sync and Azure Backup.
- A powerful way of creating a hybrid cloud configuration: maintain servers in both environments
- Can also be used to help ease a transition from a traditional datacenter into a cloud-only mentality

Enabling quick server rollouts with Sysprep

- An easier and faster way to roll out new servers as long as you are building them all from a relatively similar hardware platform
- Save you quite a bit of time on new server builds
- Sysprep (Microsoft System Preparation Tool): A tool to prepare your system for duplication
- Allows you to create a master image of your server to reuse as many times as you want in order to roll out additional servers
- A key benefit to using Sysprep: can put customized settings onto your master server and install things such as Windows Update prior to Sysprep

Enabling quick server rollouts with Sysprep

- Why Sysprep is even necessary?
- Could we simply use a hard disk imaging tool or copy and paste the .VHDX file itself in order to make a copy of your new server
- an exact replica of the original one: hostname, Security Identifier (SID) number, would be exactly the same => cause conflicts and collisions
- Each system within your network has a unique SID/GUID—their identifier within Active Directory
- Sysprep fixes all of these inherent problems with the system duplication process by randomizing the unique identifiers in the operating system

Enabling quick server rollouts with Sysprep

The steps to take

1. Install Windows Server 2022 onto a new server
2. Configure customizations and updates onto your new server
3. Run Sysprep to prepare and shut down your master server
4. Create your master image of the drive
5. Build new servers using copies of the master image

Enabling quick server rollouts with Sysprep

The steps to take

1. Install Windows Server 2022 onto a new server
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5. Build new servers using copies of the master image



Enabling quick server rollouts with Sysprep

Install Windows Server 2022 onto a new server

- Prepare the master server by getting the Windows Server 2022 OS installed
- Not to install roles onto the master server
- Install OS and make sure device drivers all squared away
- And you're ready for the next step



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Enabling quick server rollouts with Sysprep

Configuring customizations and updates onto your new server

- Configure Customizations:
 - start or stop services
 - Put things such as files and folders on all of your servers, such as an Admin Tools folder, ..
- Windows Update install: Put any patches on this new server
- Help to save all of that download and installation time for each new server that you spin up

Enabling quick server rollouts with Sysprep

Running Sysprep to prepare and shut down your master server

- Open up an administrative Command Prompt and browse to C:\Windows\System32\Sysprep
- Run sysprep.exe

System Preparation Tool 3.14

System Preparation Tool (Sysprep) prepares the machine for hardware independence and cleanup.

System Cleanup Action

Enter System Out-of-Box Experience (OOBE) ▾

☒ Generalize

Shutdown Options

Shutdown ▾

OK Cancel

Enabling quick server rollouts with Sysprep

Running Sysprep to prepare and shut down your master server

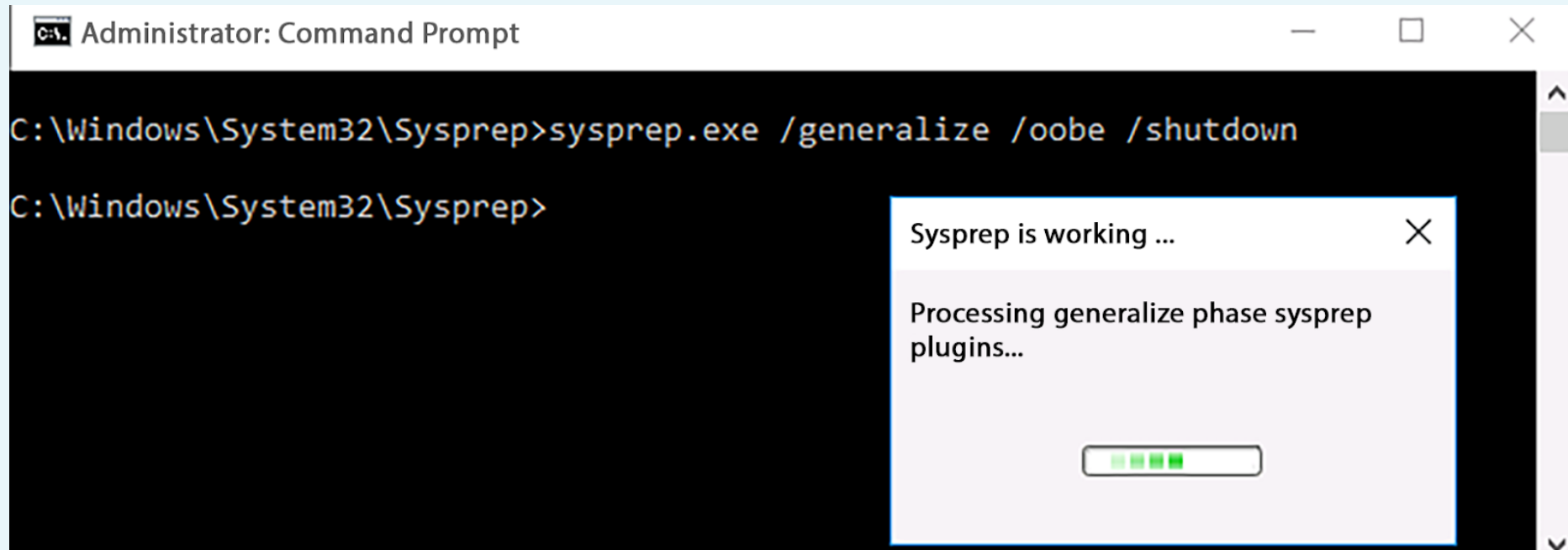
Sysprep.exe options:

- /quiet: without status messages on the screen
- /generalize: To remove all of the unique system information (SID) from the Windows installation (each new server get a new, unique SID).
- /audit: Restarts the machine into a special audit mode for adding additional drivers
- /oobe: Launch the mini-setup wizard when Windows next boots
- /reboot: Restarts when Sysprep finished
- /shutdown: Shuts down the system (not a restart) when Sysprep finished
- /quit: closes Sysprep after it finishes
- /unattend: a special answer file to configure new servers as they come online

Enabling quick server rollouts with Sysprep

Running Sysprep to prepare and shut down your master server

sysprep.exe /generalize /oobe /shutdown



- Make sure NOT boot into Windows again until after you have created master image or taken master copy of the .VHDX file.



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Enabling quick server rollouts with Sysprep

Creating your master image of the drive

- Master server now shut down, and ready to create the master image
- For physical server:
 - ✓ Use any hard disk imaging utility to create an image file from the drive
- For virtual world:
 - ✓ Simply create a copy of the .VHDX file.
 - ✓ Log in to your Hyper-V Server, copy and paste the hard disk file
 - ✓ Renamed the.VHDX file to help you keep track of the current status of this image file.
 - ✓ Copy of the .VHDX file somewhere safe

Enabling quick server rollouts with Sysprep

Building new servers using copies of the master image

- When you want to create new servers:
 - Copy and paste the master file into a new location for the new server
 - Rename the drive file to something appropriate
 - And boot your new virtual machine from it
- As the new servers turn on for the first time and boot into Windows, they will run through the out-of-box experience, mini-setup wizard.
- Also, in the background, the OS gives itself a new random and unique hostname and SID information to be sure no conflicts on the network

Enabling quick server rollouts with Sysprep

Building new servers using copies of the master image

Change the server's name



PROPERTIES

For WIN-D118V6AHD2C

Computer name

WIN-D118V6AHD2C

Workgroup

WORKGROUP

In-place upgrading to Windows Server 2022

- For so many years, it was normal that in-place upgrades were almost guaranteed to tank your server, causing it to be useless in the end
- Due to this, server upgrades almost always meant replacing the server rather than upgrading it
 - Leave the production server running
 - Spin up a new one in parallel
 - Configure all of the roles and specs needed on that server
 - Migration of data and apps to the new server (hopefully, everything up and working on the new server)
 - Then, shut the old server down. This is normal; this is Windows Server upgrades.

In-place upgrading to Windows Server 2022

Things have changed...

- From Server 2012R2 to Server 2016, it was suddenly possible to be successful with an in-place Windows Server upgrade.
- Not all the time, but sometimes. Since it was still sporadic, it was not commonly attempted, and we continued to stick to old methods of replacing servers.
- Server 2019 came out, and in-place upgrades from 2016 to 2019 were actually quite stable
- Windows Server 2022 continues that trend.
- Windows Server operating system in-place upgrades are stable and introduce us to a much more efficient way to upgrade servers



Using Windows PowerShell

- Modern replacement for the MS-DOS shell
- Work with Windows PowerShell:
 - Some key commands use to perform system administration
 - How to query system information using WMI
 - The creation and usage of PowerShell scripts.



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Using Windows PowerShell

Working with Windows PowerShe

- Right-click the Windows PowerShell icon on the Start menu and choose More, Run as administrator.
- After Windows PowerShell start, you are placed in your home directory on the system and given a PS prompt, where you can type cmdlets
- If log into system as Administrator, you will receive the following prompt:

PS C:\Users\Administrator>

- Windows PowerShell used to execute most MS-DOS commands, as well as many UNIX/Linux commands.
- For example, PS C:\Users\Administrator>copy C:\Windows\System32\calc.exe C:\



Using Windows PowerShell

Working with Windows PowerShell

- Support MS-DOS shell features: output redirection (>>) and command chaining (;)
- For example:

```
PS C:\Users\Administrator>ipconfig >> C:\IPconfig.txt ; cls
```

save the output to a file called C:\IPconfig.txt, and then clear the screen
- Commands in PowerShell called cmdlets
- Each cmdlet has an action-object or verb-noun structure.
- For example:
 - Get-Host: “get” (verb) information about “host” computer (noun).
- Cmdlets are case-insensitive, so Get-Host and get-host are equivalent.



Using Windows PowerShell

Working with Windows PowerShell

- Some examples of common cmdlets: Get-Process, Set-Date, and Write-Error.
- To see a list of all cmdlets within Windows PowerShell, run the command:

```
PS C:\Users\Administrator>Get-Command | more
```
- Pipe symbol (|) sends the results of the Get-Command cmdlet to the MS-DOS more command
- **more** command displays the output page-by-page
- pipe symbol: one of the most useful ways of sending information between cmdlets to build more complex commands or filter output to display only the output that they wish to see.
- This process called piping.



Using Windows PowerShell

Working with Windows PowerShell

- Cmdlets may also have several options:
 - available to control their execution
 - accept various arguments indicate the objects need to process.
- For example,
PS C:\Users\Administrator>Get-Process | more
lists all processes on the system
- Use the Get-Help cmdlet to list the usage of the Get-Process cmdlet (including available options and arguments)
PS C:\Users\Administrator> Get-Help Get-Process



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Using Windows PowerShell

System Administration Commands

- Some key PowerShell cmdlets and commands that administrators often use to manage Windows Server 2019.
 - Restart-Computer
 - Stop-Computer
 - Rename-Computer newcomputername: to rename a computer
 - Add-Computer –DomainName domainname: to join it to an Active Directory.
 - Get-WindowsFeature | ogv: display installed and available roles and features
 - Install-WindowsFeature: install roles and features



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Using Windows PowerShell

System Administration Commands

- For example:
 - `Install-WindowsFeature –IncludeAllSubfeature –IncludeManagementTools web-server`: Install IIS Web server
 - `Remove-WindowsFeature web-server`: Remove Web server
- Windows PowerShell also used to configure and troubleshoot the network
 - `Test-NetConnection`: Test a connection to identify problems with IP configuration, firewall configuration, and name resolution.
 - `Test-NetConnection triosdevelopers.com –Port 443`
 - `Test-NetConnection triosdevelopers.com –Traceroute`



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Using Windows PowerShell

System Administration Commands

- For example:
 - `Install-WindowsFeature –IncludeAllSubfeature –IncludeManagementTools web-server`: Install IIS Web server
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Using Windows PowerShell

System Administration Commands

- Configure and troubleshoot the network
 - Get-NetIPConfiguration: obtain information about IP configuration
 - Get-NetAdapter: network interfaces
 - Get-NetAdapterStatistics cmdlets: network statistics
- To configure a network interface called Ethernet with an IPv4 address of 192.168.1.50, 24-bit subnet mask, default gateway of 192.168.1.1, and preferred DNS server of 8.8.8.8, you could run the commands:

New-NetIPAddress –InterfaceAlias Ethernet –IPAddress 192.168.1.50 –
PrefixLength 24 –DefaultGateway 192.168.1.1

Set-DNSClientServerAddress –InterfaceAlias Ethernet –ServerAddresses 8.8.8.8



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Using Windows PowerShell

Using WMI within Windows PowerShell

- Management Instrumentation (WMI): allow programs and system software to query the hardware and software on the Windows computer
- WMI infrastructure consists of three components that work together:
 - WMI namespaces
 - WMI providers
 - WMI classes



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Using Windows PowerShell

Using WMI within Windows PowerShell

- WMI namespaces: represent/organize different types of WMI data in much the same way that a file cabinet organizes files.
- different namespaces for different purposes, and each namespace holds different WMI providers
- Only namespace useful to IT administrators within Windows PowerShell is the CIMv2 namespace (Common Information Model version 2)
 - query the hardware and software components on systems
 - modify software components as necessary



Using Windows PowerShell

Using WMI within Windows PowerShell

- WMI providers: like the drawers in a file cabinet.
 - Each provider contains different WMI classes used to obtain/modify different hardware and software information on your system.
- WMI classes: individual types of data and like the files in each drawer of a filing cabinet.
 - It is these classes that we can use within Windows PowerShell to work with different pieces of hardware and software.



Using Windows PowerShell

Using WMI within Windows PowerShell

- Query and manipulate WMI classes

Get-WmiObject (alias gwmi) cmdlet

- For example, the following command queries the win32_bios class in the CIMv2 namespace on the local computer (“.”):

```
gwmi -class win32_bios -computer "." -namespace "root\cimv2"
```

- By default, PowerShell assumes the local computer and the CIMv2 namespace. So, the following Windows PowerShell command equivalent to the previous one:

```
PS C:\Users\Administrator>gwmi win32_bios
```



Using Windows PowerShell

Using WMI within Windows PowerShell

- Instead of specifying the WMI class with Get-WmiObject cmdlet, we can instead create a WMI query statement and specify it using the –query option.
- WMI query statements use a SQL database language called WQL (WMI Query Language).
- For example, the following command within Windows PowerShell will select all attributes/ methods (*) from the win32_share class as shown below:

```
PS C:\Users\Administrator>gwmi –query "select * from win32_share"
```



Using Windows PowerShell

Creating PowerShell Scripts

- PowerShell scripts simply text files with a .ps1 extension executed within PowerShell
- PowerShell scripts can contain both Windows commands and PowerShell cmdlets, as well as complex control structures.
- The script can be used to accept input that changes how the script works.
- The use of variables and constants, protecting metacharacters, coloring and formatting output, and the use of decision and loop constructs.



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Using Windows PowerShell

Creating PowerShell Scripts: Executing PowerShell Scripts

- Enable script support in Windows PowerShell:
 - Set-ExecutionPolicy unrestricted
- To see current execution policy: run the Get-ExecutionPolicy cmdlet.
- Five different execution policies on a Windows system:
 - Restricted: not run any scripts (script execution is disabled).
 - AllSigned: only runs scripts that are signed by a trusted publisher.
 - RemoteSigned: allows to run scripts that are downloaded from the Internet only if they are signed by a trusted publisher.
 - *Unrestricted*: runs all scripts (scripts downloaded from the Internet prompt for confirmation before execution).
 - *Bypass*: runs all scripts (no warnings).
 - *Undefined*: indicates that a policy is not set.



Using Windows PowerShell

Creating PowerShell Scripts: Executing PowerShell Scripts

- To execute a PowerShell script, use a full path or a “forced” relative path (where “.” refers to the current directory):
 - Full path: C:\scripts\myscript.ps1
 - Relative path (within the C:\scripts directory): .\myscript.ps1 or ./myscript.ps1
- To execute a PowerShell script outside Windows PowerShell, use the powershell
- For Example:

`powershell C:\scripts\myscript.ps1 –noexit`

–noexit prevents Windows PowerShell from closing after executing the script)

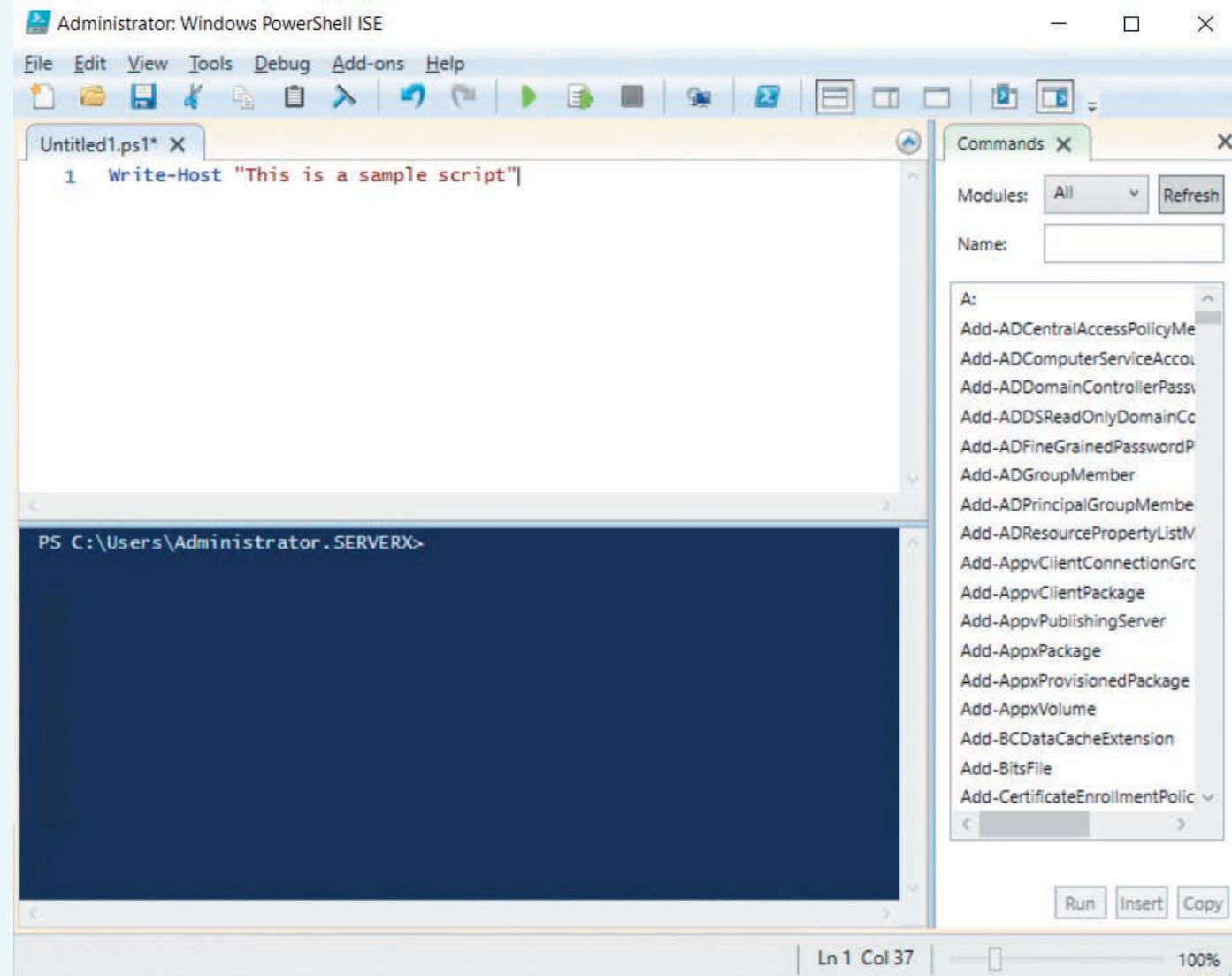


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Using Windows PowerShell

Creating PowerShell Scripts: Using Windows PowerShell ISE

- Windows PowerShell Integrated Scripting Environment (ISE): tool for creating, testing, and executing scripts
- Windows PowerShell pane on the left
- Commands pane on the right allows to search for and execute commands
- click the Script opens a new PowerShell script in the upper pane called Untitled.ps1





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Using Windows PowerShell

Creating PowerShell Scripts: Using Windows PowerShell ISE

- After adding content to Untitled.ps1, clicking the green Run script (play button) icon or by pressing the F5 key to execute it
- File -> Save As, and choose to save it with a more descriptive file name to location of your choice (e.g., C:\scripts\myscript.ps1).
- You can also create and test scripts on remote computers that have winRM started within Windows PowerShell ISE.
- To do this, navigate to File, New Remote PowerShell Tab and specify the correct computer name.
- If you are currently logged into an Active Directory domain with domain administrative privileges, you will be allowed to connect to any computer within your domain that has winRM started.

Summary

- Requirements for installation
- Installing Windows Server 2022
- Installing roles and features
- Centralized management and monitoring
- Windows Admin Center (WAC)
- Enabling quick server rollouts with Sysprep
- In-place upgrading to Windows Server 2022